AI in High-Tech Manufacturing: Firm-Level Evidence from a Multi-Plant Rollout *

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Abstract

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1 Introduction

Recent advances in Artificial Intelligence (AI) systems promise to radically alter the workplace, affecting labour demand through the automation and augmentation of tasks [1]. Human level performance has been met or exceeded across a wide range of benchmarked tests, including image classification, reading comprehension, and language understanding, though performance gaps persist on more demanding reasoning and domain-expert benchmarks[2]. Although benchmark saturation does not necessarily map onto real-world task competence, various forms of occupational exposure predict significant substitution, assigning large shares of employment to tasks that AI can in principle perform [3, 4, 5]. Automation technologies substitute for labour at the task level, leading to reduced labour share, shifting occupational structures and reducing employment for workers in automation exposed occupations [6]. However, the realized employment effects of AI have so far been modest. Studies relying on task-based occupational exposure scores [7, 4, 3], skills-based vacancy measures [8, 9, 10, 11] and firm- or worker-level survey data [12, 13, 14, 15, 16] generally report small or null average effects on employment and wages.

Two explanations can account for the distance between predicted exposure and realized effects. First, automation generates countervailing forces. If the productivity effect is sufficiently large, it can counteract any displacement, minimising labour share reductions[1]. Offsetting can occur through increased output, or through the reallocation of worker effort, where workers endowed with the relevant skills redirect effort toward unaffected tasks [17]. Second, a more fundamental constraint is that realized displacement is bounded by the pace and extent of diffusion processes. Occupational exposure scores capture the substitution *potential* implied by current technological capability, but the labour-market consequences of AI depend on whether, and how quickly, firms integrate these tools into production. Even where reallocation across tasks is feasible, rapid diffusion may not allow workers sufficient time to adjust [18]. Thus, the pace of technologically diffusion could impact labour demand outcomes.

Outside of Generative AI tools[1] [13, 14], the diffusion of AI technologies has been limited and heterogeneous across sectors, dominated by larger and more productive firms [16]. The study of AI diffusion and it's consequences has predominantly been carried out using firm level events, such as announcements on firm websites [22, 23] or adoption events captured in a survey [24, 16]. Yet a firm-level adoption event is only part of the broader process of intra-firm diffusion, which typically proceeds more slowly than inter-firm diffusion [25, 26]. Previous studies revealed how the intra-firm diffusion of older technological vintages, such as CNC machines [25], IT technologies[27, 28], e-commerce [29], and ATM machines [26], often unfolds more slowly than inter-firm diffusion. The process of intra-firm diffusion has largely been shown to be driven by rank effects, rather than epidemic learning diffusion or stock/order effects. Thus, it is moderated by the capabilities of the adopting unit. However, little evidence has been provided on whether this pattern holds for AI tools. As they are prediction machines, requiring significant technical know-how and trust, learning effects from epidemic models may predominate [23]. Other factors such as algorithmic aversion may alter diffusion processes compared to previous technological vintages, where decision making power has not been automated [30].

Most studies of diffusion measure adoption at the unit level, recording whether a technology is used within an establishment or by an individual. Returns to adoption depend not only on the extensive margin, but also on the intensity with which the technology is used once introduced [29]. Separating the two margins requires observing both whether a technology has been enabled and how much of the work it performs once enabled, which unit-level and cross-sectional adoption measures often cannot deliver.

[1] One can question the extent to which the diffusion of Generative AI tools is relevant to the diffusion of traditional machine learning (ML) AI tools, based on pre-trained models designed for specific tasks. Generative AI adoption so far has been worker led, demonstrating a lack of formality [12, 19, 20]. While traditional ML tools have typically been more top-down with formal firm led adoption involving make or buy decisions. [16] The faster pace of Generative AI adoption may help explain the emerging evidence for displacement among junior workers [21]

Our unique data allows us observe both the extensive margin of adoption and the intensive margin of use as measured by task reallocation, and we can therefore establish whether the growth of AI use within a firm comes from expansion across units or increased use within units.

The objective of this paper is to provide descriptive evidence on the pace and extent of AI diffusion within a multinational electronics manufacturer, linking diffusion processes to task reallocation and shedding light on shifts in labour demand. We use proprietary data from the deployment of a machine learning-based AI visual inspection tool for printed circuit board (PCB) assembly quality assurance (QA) across 14 plants between September 2022 and November 2023. PCB quality assurance is a well-defined, high-frequency decision task which is highly suitable for machine learning across a number of dimensions including: large, well-labelled datasets; well-defined inputs and outputs, goals and feedback; no specialised dexterity or mobility required; and task stability [31]. As prediction machines, AI tools are well placed to automate such decision tasks [32].

Given this, one might even expect full automation of the decision task by AI, however, the firm is a supplier to the automotive industry and many of the PCBs it produces are used for safety critical applications, such as anti-lock brake systems and airbag deployment, requiring exacting safety standards. It has previously been suggested that diffusion processes will be slow in safety critical applications for AI [33]. Beyond high safety standards, malfunctioning electronics can lead to production recalls and significant financial and reputational damage. This limited tolerance for error suppresses full automation of the visual inspection task. While the decision to deploy the AI tool was taken by the firm's central AI and data science team, plants had considerable discretion in choosing when they could begin deployment and the extent to which the AI was used for the inspection.

Our decision data is highly granular, capturing the final decision, on which electronic component (i.e., transistor, resistor) the decision was made, the PCB the component is soldered on, the date the decision was made and whether the decision was made by a human or the AI tool. We begin with descriptive analysis mapping the diffusion of the AI tool on components, across different PCB products, over plants. We leverage the date at which plants began deployment to investigate differences between early and late adopters. Following this, we exploit the share of decisions assigned either to the operative or the AI to investigate the relationship between automation technology and labour displacement within this task. We combine this with qualitative insights gather unstructured discussions with key decision makers within the company.

We document two findings. First, AI use expands almost entirely along the extensive margin. The share of decisions made on components where the AI has been deployed rises from 30% to 70% over our observation window. At the intensive margin the AI automates around one third of decisions once deployed on a product, and we detect no systematic growth in that share over the observation period. Overall, the share of decisions the AI actually executes rises from 16% to 27%, proportionally reflecting the expansion along the extensive margin. Plants therefore expand automation by deploying the AI on additional products and component libraries rather than by raising the share of decisions it takes within existing deployments. Splitting plants by deployment date, early adopters have higher automation levels at both margins, while late adopters exhibit faster relative growth at the extensive margin once the technology is introduced, albeit with substantial heterogeneity. Applying theories of technology diffusion [34], and combining our empirical results with firm-specific insights our findings seem to fit more closely with rank effects rather than epidemic models.

Second, the AI absorbs inspection decisions without enlarging the inspection task. We find no relationship between changes in the AI Automation Rate and the total volume of QA decisions, so the countervailing productivity effect emphasized in the automation literature is absent in this setting. Operator decisions fall as automation rises, at a rate statistically indistinguishable from the one implied by the accounting relationship between the two measures, which indicates a near one-to-one displacement

process. Because we observe decisions rather than workers, we cannot establish whether that transfer translates into changes in operator hours or employment. We also observe a small increase in the overall pass rate, consistent with fewer non-defective components being incorrectly flagged as defective, indicating improved prediction and decision making [32]. Therefore, within the QA task, the AI changed who takes the decisions rather than how many decisions are taken. Taken together, these results demonstrate that the nature of the diffusion process is vital for task reallocation and labour displacement.

The remainder of this paper is organized as follows: in Section 2 we discuss the context of the paper, explaining more about the firm and setting we study. In Section 3 we describe the data we used and any relevant data cleaning process. In Section 4 we describe the diffusion processes of the AI tool through the firm and it's relationship to labour demand. Section 6 concludes.

2 Empirical Context

2.1 Company Setting

We evaluate the the effects of AI adoption in a large a global international electronics manufacturer who develop, supply and produce technologically advanced products and services for original equipment manufacturers in the automobile industry. The company is one of the largest in its market, with a global reach, around 100,000 employees, and sales figures in the tens of billions of euros. Despite this, competitive pressures are high, both from traditional competitors and new comers into the market. This pressure demands the company invest in continuous innovation and process improvement to maintain their market position. The successful adoption of AI is viewed by the company as crucial step in forward planning.

Specifically, we evaluate the adoption of AI in a particular task within this company. We focus on a QA task that involves inspecting soldering points on Printed Circuit Boards (PCBs) to detect anomalies. Performed by quality control workers, this task is particularly demanding, requiring sustained visual attention and precision to identify minor defects at high production volumes. In what follows, we provide details on the company's production process, the nature of the QA task before AI, and the subsequent introduction and deployment of AI technologies that automate parts of this process.

2.2 Printed Circuit Board Assembly

The company assembles PCBs using surface-mount technology (SMT), a highly automated and technologically advanced production method with widespread use in PCB assembly. An exemplary representation of a circuit board and the SMT production environment is shown in Figure 1. In SMT production lines automated pick-and-place machines put electronic components—such as capacitors, diodes, resistors and transistors—onto circuit boards that have had solder paste painted on at specific solder points. These boards are then passed through a reflow oven, melting the solder and permanently connecting the components to the board. For a more detailed overview of the process, see [35].

Following the reflow soldering process, the assembled PCBs undergo visual inspection to detect manufacturing defects, such as faulty soldering points or misaligned components. As PCBs are essential for the correct functioning of all electronic devices, flawless performance is essential. Even minor defects, such as micro-cracks in solder joints or insufficient soldering, can lead to functional failures. In many of the company's end products, which include safety-critical applications (e.g., automotive safety and control systems), such failures can have severe consequences. As the SMT production process allows for smaller component sizes and higher circuit densities than alternative assembly processes (such as through-hole assembly), there is an increased risk of defective units. As a result, the QA process is a critical stage in the production process, crucial for identifying even minor deviations from manufacturing standards [36]. This ensures product reliability, operational safety, and compliance with stringent industry regulations and customer demands. In our case, customers impose particularly high accuracy requirements and detailed inspection protocols, ultimately shaping how extensively the AI system can be utilised in the QA task.

2.3 QA before AI

Prior to the introduction of AI, the visual inspection task of PCBs was carried out by human quality control operators and a rule-based Automated Optical Inspection (AOI) system. AOI systems capture high-resolution, multi-angle images of PCBs and compare them to a predefined "golden" template [37]. If the captured images match exactly, then the component is marked as non-defective (or OK) and continues

in production. If there are any detected deviations, components are flagged as defective. Because the rule-based algorithms operate with limited flexibility, they tend to generate a substantial share of false negatives, flagging components as potentially defective even when they conform to production standards.[2] Since defective components must be either scrapped or manually repaired, this high false negative rate would introduce significant waste without further inspection. Human operators therefore play a vital role in improving the efficiency of the process: they review all components flagged by the AOI as defective, reclassifying OK components while catching any truly defective ones that must be removed from production.

Operators are selected from unskilled production line workers called assemblers, whose primary task is to load materials into the SMT production line. Assemblers can apply to become operators, though the position offers no pay premium and provides only the benefit of being seated during the task. The visual inspection task itself is cognitively demanding, requiring sustained visual attention and precision to identify the few defects within a high volume of non-defective images. Of the images seen by operators, 90-95% are eventually classified as OK. Due to these demand, workers must pass a simulation-based assessment at the beginning of each shift before being allowed to do the inspection task. Furthermore, operators are only assigned to this task for a portion of their shift, typically around 2 hours, after which they return to regular assembly duties.

2.4 AI-Automated QA Decisions

The traditional QA workflow, while effective at catching defects, remains inefficient: due to stringent quality requirements, operators are trained to err on the side of caution. This results in a higher scrap rates than necessary. Developments in machine learning technologies created an opportunity for the company to tackle this inefficiency, while simultaneously reducing labour costs without compromising on quality. Based on an algorithm design taken from a Google white paper, the company's AI and data science division developed a supervised learning classification algorithm for the QA task.

The AI system is designed to reduce the volume of images requiring human review by acting as an intermediate classifier between the AOI and human operators. Positioned after the rule-based AOI, the AI reviews components flagged as defective and autonomously reclassifies those it can confidently predict to be OK, allowing them to continue in production without operator review. These are actual share of AI-automated decisions as previously they would have required an operator to classify them as OK or defective. To ensure ongoing quality control, a sample of AI-classified OK components are also sent to operators for verification. Components that the AI flags as defective, or cannot classify with sufficient confidence, are forwarded to operators for further inspection (see Figure 2 for an overview of this process). This changes the nature of the operator's task: rather than reviewing the full set of AOI-flagged components, operators now handle a filtered subset consisting primarily of ambiguous or defective cases, alongside a small subset of OK decisions for monitoring.

2.5 Training and Deployment of the AI

The company's AI and data science division developed the algorithm, training procedures, and deployment pathways at a global level. Actual training and deployment, however, occur at the plant level due to differing production technologies across plants. Therefore, the AI is trained for each plant using a locally curated test dataset containing samples of regular production data, as well as sufficient numbers

[2]We define false negatives as components flagged as defective that are actually OK. True negatives are components flagged as defective that are actually defective. True positives are components flagged as OK that are OK, false positives are components flagged as OK that are actually defective. Note that in quality assurance literature, this terminology is often inverted, with 'positive' referring to defect detection rather than non-defective classification, e.g., [37].

of images of each known error class. These images are then labelled by certified PCB experts before being used to train the AI. While operator-generated labels can be incorporated into the training process, they are typically included only after being verified by experts, and the extent to which they are used depends on the maturity of the model at the respective plant.

The images used for AI training are grouped based on the component captured. These groupings are referred to as component libraries, where a component library contains physically similar components sharing common visual features (e.g., all resistors of a specific size). Training is entirely supervised and component-specific, with no ongoing learning after deployment. Before deployment, the AI must achieve a false positive rate under 2%, and a false negative rate under 10%. As all components flagged as defective are eventually reviewed by operators, there is higher tolerance for false negatives. The false positive rate offers a substantial improvement over human operators, who are estimated by the company to have rates as high as 10 or 15%. Once the AI has exceeded these thresholds, plants can begin the deployment process.

Deployment occurs through a multi-step, gradual process contingent on the technical readiness of the AI model, customer approval, plant confidence in the technology, and plant organisational readiness. As a result, even highly capable AI models are not deployed uniformly or at full scale. This process proceeds along two margins: extensive and intensive. At the extensive margin, the AI is deployed on new products and component libraries. Once a model is trained and approved for release by the plant, it cannot be used on any product without customer approval. The company must therefore convince the customer that the AI provides a substantial quality improvement over the traditional workflow.

Once the AI has been deployed on a component, plants can adjust the intensive margin through several mechanisms. They may lower certainty thresholds over time, allowing the AI to handle a larger share of decisions. They may retrain models with updated data, improving the AI's confidence on previously marginal cases. Finally, they may reduce the sampling rate for operator verification of AI-approved decisions. These adjustments vary considerably across plants, resulting in substantial heterogeneity in how intensively the AI is utilized. We explore adjustments at either the extensive or intensive margin further down, in Section 4.1.

2.6 Role of human operators after AI deployment

Following AI deployment, human operators continue to play a crucial role in the QA process. They review all components for which no AI model has been trained, handle cases where the AI cannot decide with sufficient confidence, and inspect all components the AI flags as defective. Additionally, operators verify a sample of OK decisions made by the AI, allowing plants to monitor the AI's accuracy over time.

This changing role has significant potential implications for the work of operators. With a fully deployed AI, their task composition changes as they see a higher proportion of defective decisions, and potentially more challenging components to classify. They also become an important mechanism for ensuring the AI continues to operate acceptably. Further, they are exposed to decreasing work demand in the verification task due to task replacement. This could either free up time for them to spend on the production line as assemblers, or reduce the overall demand for assemblers within the plant, potentially leading to job losses.

In the next section we investigate the relationship between AI deployment on the QA process through the analysis of decision data provided to us by the company. We focus on changes in labour demand within the QA task, total decision throughput, and the outcomes of the QA process to elicit insights into the changing nature of the QA task.

3 Data

3.1 Data Overview

The company provided us with access to unique QA decision data compiled specifically for this research project. The data cover 16 plants operating in 8 countries across Asia, the Americas, and Europe, and include the date, time, and result of all QA decisions made by human operators and the AI. The data also include anonymized identifiers for the components inspected, the products on which they appear, and their associated component libraries.

QA decisions are made on images of individual components. Each component belongs to both a specific product (the PCB being inspected) and a component library (a grouping of similar components, as described in Section 2.5). The relationship between these levels is illustrated in Figure 3. Our primary analyses use a week-plant panel, though we also construct a week-component library panel for robustness checks. The relationship between products, component libraries, and components is illustrated in Figure 3. Our primary analyses are conducted using a week-plant panel, and we also construct a panel at the week-component library level for additional analyses.

Our main variable of interest is the **AI Automation Rate**. This is defined as the share of QA decisions actually automated by AI relative to the total number of QA decisions. Automated decisions are those that are independently classified as "OK" by the AI without any operator review.[3] This measure captures the intensity of AI-driven automation in the QA process across plants and over time, reflecting adjustments at both the extensive and intensive margins, and forms the basis for the analyses presented in the subsequent sections. Since every QA decision is taken either autonomously by the AI or by a human operator, the AI Automation Rate and the volume of operator decisions are constructed from a common total. We return to the implications of this in Section 4.2.2. Our focus on "OK" decisions follows directly from how the AI system operates within the QA workflow. As described in Section 2.4, the AI is only allowed to make an autonomous decision when it classifies an image as non-defective. These cases constitute the part of the inspection task that the system can execute independently, without human verification. By contrast, any image the AI flags as defective, or is uncertain about, is routed to the human operator for review. We do not focus on the AI defect rate as the thresholds for the AI to make a decision are set significantly lower, and thus operators overturn up to 75% of AI defect decisions. Measuring the share of AI-approved OK decisions relative to all QA decisions therefore provides a direct indicator of the portion of the QA task that the AI can execute without human review.

Using this automation variable as a starting point, we introduce several measures in the following sections that allow us to (i) characterise how AI deployment unfolds over time and across plants, and (ii) examine how changes in automation intensity relate to labour demand within the QA task.

3.2 Sample Selection and Summary Statistics

We implement several sample restrictions to ensure data quality and consistency. First, we exclude two plants that contained no information on operator decisions, leaving 14 plants for analysis. Second, we address a data reporting issue that emerged toward the end of our observation period: across multiple plants, operator decision data were not recorded despite ongoing operator activity. To mitigate potential measurement error from this reporting gap, we truncate the sample, removing all observations from the affected period. This restriction eliminates 27.7% of plant-week observations, corresponding to 22.4% of total decisions. Our final estimation sample comprises 14 plants observed over 60 weeks, from September 2022 through November 2023. This approach helps us to account for a systematic data collection problem

[3] Section 2.4 provides a detailed description of the AI system and the automated decision making process.

across plants, however, there remain persistent issues with the data throughout our observation period relating to missing product and decision information.

Within our final sample, per week on average 9% of decisions are missing product and component identifiers. Thus we cannot assign them into component libraries or identify the products they were made on. Missing product information occurs across all plants, throughout the observation period. Decisions with products missing have a 21% lower AI Automation Rate than non-missing, and are 5% less likely to be passing decisions. For our primary analysis, we remove those decisions with missing product information. This is likely to overestimate the AI Automation Rate, therefore we run robustness checks to test the sensitivity of our results to missing product information by including those decisions in our final total. We also run sensitivity analyses using our variable construction, but excluding weeks with high levels of missing product information. A further data issue arises from lines in the data missing decisions information, but containing product identifiers. On average, around 6% of data rows per week contain no decision information but have an assigned product identifier. To account for this, we identify weeks with a high level of missing product information and in a sensitivity analysis remove them from our sample.

Table 1 summarizes key characteristics of these variables across 14 plants and 60 weeks during the observation period in the plant-week panel. In Panel A, we observe that most plants have the full set of 60 week observations. However two plants have missing data resulting in only 53 and 57 week observations, thus averaging 59 week observations for plants. Plants make decisions across an average portfolio of 902 different products and 1,118 component libraries during the observation period. Plant size varies substantially: the smallest plant accounts for only under 0.2% of total QA decisions whereas the largest contributes to 42%. This heterogeneity in plant sizes presents challenges for our empirically strategy. On the one hand, we want to avoid results being driven disproportionately by the largest plants; on the other hand, we want to prevent very small plants having an outsized influence. To mitigate these concerns, we express outcomes in logs where appropriate and report both unweighted and weighted regression results.

Table 1 Panel B reports on the average AI Automation Rates and QA Outcomes. We find that the lowest plant has an average weekly AI Automation Rate of only 1.8%, whilst the highest plant has 67.9%. The average AI Automation Rate across plants is 23.1%, thus on across plants roughly 75% of decisions are taken by human operators. For all QA decisions, irregardless of whether they have been taken by the AI or a human operator, we find a high average OK rate of 95.3%, with some variation between plants ranging from 86.7% to 99.8%. The Operator OK Rate is slightly lower, at 93.8%. This discrepancy suggests that the AI influences the OK rate at the plant level. Operators could be more likely than the AI to flag a component as defective,alternatively this could reflect changes in task reallocation with Operators handling a higher share of defective images as the AI automates only OK images, we explore the this in Section 4.4.

To assess the impact of AI Automation, we rely on weekly first differences, therefore in Table 1 Panel C, we report on week-to-week changes in QA decisions. We first report on our explanatory variable, AI Automation Rate. We see an average weekly growth in the AI Automation Rate of 0.164 percentage points across plants, across our observation window. Turning to Total Throughput, defined as the total number of decisions taken by operators and AI, in general we observe small, positive average weekly changes of about 0.8%, suggesting that the plants increase there throughput during our observation window. Turning to Total Operator Decisions, we similarly see a slight increase of 0.6%. Whilst similar in magnitude, albeit slightly slower, providing an indication that there is substitution between labour and capital within this task. Measuring the increase in AI decisions, we see that the Total AI Decisions increases by about 3% per week, a higher growth rate than either Total QA Throughput or Total Operator Decisions. The above descriptive evidence suggests that the increase in decisions being assigned to AI is outstripping increases in total throughput, leading to a widening gap between productivity increases

and labour demand.

4 Results

In this section, we first document how AI deployment unfolds across plants during our observation period, highlighting differences between early and late adopters and the relative contributions of the extensive and intensive margins. We then turn to labour demand, where we assess how week-to-week changes in the intensity of AI automation relate to changes in operator activity at the plant level and the empirical strategy guiding our analysis. Finally, we examine two additional margins of adjustment, overall QA volume and QA accuracy, to understand whether AI adoption affects productivity or the quality of the inspection process.

4.1 The Diffusion of AI

We start by exploring the diffusion of AI deployment across plants during our observation period using the AI Automation Rate. We have precise start dates for all plants that began deployment within 15 months before the start of our observation period, with the remainder deploying earlier. Of the fourteen plants in our sample, nine had already begun AI deployment prior to the start of the observation period, while five deployed during our observation window. Among the nine that deployed before our sample, seven began more than 8 months prior, while two began 1-4 months before. The five plants that deployed during the observation period began at various points in time: three within the first 3 months and two between 8-11 months after the sample start. Table 2 provides the full distribution of deployment timing. To capture heterogeneity in deployment phases, we classify plants as "early adopters" if they began deployment more than 8 months before our sample period (N=7), and as "late adopters" if they began deployment 8 months or less before the sample start or during the observation window (N=7). The company informed us that technological and organisational readiness, alongside plant manager characteristics (e.g., technology acceptance) influenced whether a plant was a late or early adopter. Competition among plants is encouraged within the firm, with some plant managers wishing to demonstrate productivity increases while others being more cautious in adopting newer technologies.

Figure 4 plots the AI Automation Rate averaged for all plants and split between early and late adopters. Figure 4a shows that the overall AI Automation Rate across plants remains relatively low, starting at around 16% at the beginning of the observation period and reaching 27% by the end. Between early and late adopters, we observe a persistent difference in levels, with early adopters maintaining higher automation rates throughout our observation period. Figure 4b compares the trajectories of early and late adopters by indexing to Week 1, revealing that both groups exhibit similar patterns of increase over time. This suggests that after initial deployment, plants continue to gradually increase automation rather than reaching an equilibrium. This indicates that AI implementation is an ongoing process rather than a discrete event. Figure A.1a in the appendix shows individual plant trajectories, confirming that the group means are representative despite variation across plants within each adopter group.

What drives this sustained automation growth? In Section 2.5 we distinguished two mechanisms: at the extensive margin, automation can increase as AI is progressively enabled on additional products or a newly trained algorithm is released on an additional component library. At the intensive margin, automation can rise as the thresholds at which the AI can make an autonomous decision are adjusted, or a retrained algorithm with additional images is deployed.

To examine the extensive margin of automation, Figure 5 plots the share of total QA decisions made on components where the AI has been deployed, which we refer to as Potential AI Automation Rate. Unlike the AI Automation Rate, this measure contains both automated decisions and those taken by

operators. Thus, it captures the expansion of AI deployment onto new products and new components but not necessarily the expansion of AI Automation. Figure 5a shows that the share rises from 30% to 70% across plants, through our observation period. Again, we observe a levels difference between early and late adopters, with a higher extensive margin for early adopters throughout. However, moving to Figure 5b, we begin to see differing patterns across the two groups. When indexed to the first week's value, we see that late adopters increased the extensive margin by 40 percentage points compared to just 20 percentage points for early adopters. Comparing the automation rate to expansions at the extensive margin, we see that the increase at the extensive margin is higher than the overall increase in the automation rate. This suggests that plants deploy the AI at a relatively low intensity and increase as time progresses.

To understand the intensity of AI use, we investigate the AI Automation Rate on components where the AI has been deployed (i.e., the intensive margin). The average AI Automation Rate once the AI has been deployed on a component is around 36% for the entire period, showing that the AI only automates around one third of decisions leaving a substantial share for operators to review even after deployment Splitting into early and late adopters, we find a rate of 44% for early adopters and 28% for late adopters suggesting that plants who began using the AI earlier use it more intensely at both the extensive and intensive margins. However, even after reaching maturity, plants are still cautious about how much to automate requiring human oversight on over half of the AI's decisions.

This pattern is illustrated in Figure 6, which plots the AI Automation Rate, removing all decisions made on components where the AI is not deployed. Figure 6a reflects the averages described above; we find that on average there is a lower *intensive* usage of the AI by early adopters compared to the late adopters. If we move to the indexed change, in Figure 6b we see that for both early and late adopters the adjustments at the intensive margin appear to be more random and show no systematic growth or decline over our observation period. The equilibrium reached at the intensive margin suggests that learning does not occur on how intensely to use the AI tool once deployed, but that diffusion occurs either through expansion onto new libraries or gaining further customer approval.

These trajectories speak to competing accounts of why intra-firm diffusion may be slow. In Epidemic models, diffusion progresses concurrently with learning. Later adopters learn from early adopters, reducing the risks of adoption and resulting in increases in adoption through time following a logistic S-curve [38]. Our growth trajectories at the extensive may indicate a mid-S-curve path: as plants acquire information about the technology they further entrench adoption, such learning is also facilitated by the firm's central AI division. However, as with previous studies of intra-firm diffusion [39], time since adoption only explains a moderate proportion (32% in our case)[4] of the variation in AI use intensity across plant-week observations. This is larger than the 10% reported in [39], suggesting that epidemic and learning effects play a larger role here than in other settings. However, much of the variance is not explained by pure time from adoption. Further, one may expect that learning effects could drive an increase in the intensive margin (i.e., as plants learn more about the AI, they trust it's performance more and adjust thresholds accordingly). However, we find no effect at the intensive margin, rather expansion is driven by the deployment on new libraries or products. If plants were Bayesian updaters, we would expect them to increase expansion on both margins. Taken together, it seems unlikely that epidemic effects entirely explains the diffusion process.

In Probit models, the decision to adopt a technology is determined not by time since adoption, but rather in rank effects. Rank effects are differences in capabilities across units, leading to different returns from adoption and hence heterogeneities in adoption [34, 40]. [39] expand the use of rank effects beyond the adoption decision, into the intensity of use of technologies within firms. Those that are

[4]Regressing usage intensity on maturity (weeks since deployment) alone, without plant fixed effects, yields an R\² of 0.322: maturity explains roughly 32% of the variation in AI decision share across plant-weeks.

larger, engage in R&D, and have better technological and managerial capabilities are more likely to have deeper technological adoption. Due to data limitations, we are unable to test these hypotheses directly. However, in discussions with the company we confirmed that there was heterogeneity across plants in their capabilities. Importantly, in our setting, rank effects within the plant themselves are not the only limiting factor. The company told us that the plants also had to individually convince their customer base to use the technology, due to it's safety critical nature. This implies that rank effects which focus purely on the unit itself and not the market that it serves are more limited in their understanding.

A further possibility concerns order effects, where returns to adoption depend on a unit's position in the adoption sequence. [39] argue that order effects do not transfer to the intra-firm setting, as any such returns are internalised within the firm. Our setting suggests this may be less clear-cut in large multinationals, where the company reports that competition between plants is actively encouraged and some plant managers sought to demonstrate productivity gains ahead of others. We lack the data to establish whether this reflects pre-emption in the sense of the diffusion literature, but it indicates that internal competition is a plausible mechanism where responsibility for adoption is decentralised.

Taken together, these figures help us to understand the average progression of AI deployment. In plants, the AI Automation Rate increases, both across early and late adopters. These changes appear to be largely driven by the extensive margin, that is the deployment of AI on new products or component libraries as we are unable to see a systematic rise in the intensive margin over our observation period.

4.2 Labour Demand

4.2.1 Empirical Approach

We now turn to examining the relationship between AI adoption and labour demand within the QA task. Two factors impede attempts at causal analysis: (1) deployment timing of the AI at product or component level is endogenous, plants are able to anticipate deployment and adjust their labour demand occurring; (2) products contain both treated and untreated components and operators work across lines with both treated and untreated components, creating spillovers that violate the Stable Unit Treatment Value Assumption (SUTVA) (see Appendix 2 for challenges to causal identification).

We therefore aggregate to the plant-level and rely on a Two-Ways Fixed Effect (TWFE) estimation in a first differenced framework to control for unobserved, time-invariant plant characteristics and estimate how week-to-week changes in the intensity of automation relate to changes in the demand for human labour. This allows us to capture the relationship between AI and labour demand throughout the plant, including any potential spillover effects on products or components where the AI had not previously been enabled.

Figure 7 plots the weekly change in AI Automation Rate against the weekly change in the total operator decisions at the plant level, indicating a negative relationship. To quantify this, we estimate the following model:

$$Y_{i,t} = \alpha + \beta \cdot \Delta\text{AI Automation Rate}_{i,t} + \gamma_t + \theta_i + \varepsilon_{i,t} \tag{1}$$

Here, $Y_{i,t}$ denotes the weekly change in the outcome of interest in plant i and week t e.g Δ Total Operator Decisions (Log). Our main explanatory variable, $\Delta\text{AI Automation}_{i,t}$, reflects the weekly change in the AI Automation Rate at the plant level. We include week fixed effects γ_t to control for common time shocks affecting all plants, such as the output growth. We also include plant fixed effects, θ_i, to account for plant specific growth trends. We cluster standard errors at the plant level. To account for the heterogeneity in plant size, we run both unweighted and weighted regressions, weighting for the total number of decisions recorded in the plant in that week.

4.2.2 Labour Demand Outcomes

We first explore the relationship between an increase in the AI Automation rate and labour demand, as proxied by the growth rate of total operator decisions (i.e. Δ Total Operator Decisions (Log)). A limitation with this approach is that we rely on the number of decisions taken by operators as a proxy for total labour demand due to a lack of data on specific operators, including total employment numbers, working hours, and time spent per decision. It could be that the total demand for operators stays the same as they are able to spend more time on the decisions that remain after automation. Discussions with the company suggest that such adjustments are unlikely to occur at scale, but we cannot observe them, and we therefore interpret our estimates as capturing the reallocation of decisions within the QA task rather than a change in employment or hours.

Moving to the results, as reported in Table 3 we find a statistically significant and economically meaningful negative relationship between weekly changes in the AI Automation Rate and the growth rate of operator decisions. The estimated coefficients, ranging from -1.201 in the unweighted specification to -1.216 in the weighted specification, imply that a 1 percentage point increase in the Automation Rate is associated with a 1.17%–1.2% lower growth rate in operator decisions. Since operator decisions equal total decisions multiplied by one minus the AI Automation Rate, the coefficients in Panel A reflect the rate of task reallocation, $1 - 1/(1 - AutomationRate)$, which is -1.31 with a mean of 23.1. Our estimates of -1.216 and -1.201 are indistinguishable from these, suggesting no countervailing productivity effect, consistent with our productivity findings in Section 4.3.

A potential concern is that plants may anticipate the introduction of AI systems and adjust their production numbers in the weeks prior to a deployment. As demonstrated in Appendix 2, there is endogeneity in deployment timing at the component library level. This could be replicated at the plant level, with reduced productivity in the weeks leading up to a large AI deployment. Thus, we may not be capturing the entirety of task reallocation. A related concern is that plants could use medium- and long-term forecast to determine AI deployment patterns, thus increases in the Automation Rate may be correlated with anticipated shifts in customer demand.

We are unable to test for medium and long term endogeneity, however we can account for the possibility of short-run endogeneity. To achieve this, we re-estimate Equation 1 by adding one- and two- week leads of Δ AI Automation Rate. In other words, we test whether the growth in operator decisions in the current week is correlated with future growth in AI Automation Rate. The results in Table 3 show no evidence of such anticipation effects: future AI deployment is not systematically related to current operator decisions. Instead, the observed decline in operator decisions coincides with contemporaneous increases in AI decision shares, suggesting that the detected relationship maps to increases in AI deployment.

4.3 Total QA Volume (i.e. Productivity)

We next explore whether changes in AI adoption affect overall plant productivity. We proxy productivity with the growth rate of total QA decisions, estimating Equation (1) using Δ Total QA Decisions (Log) as the outcome variable. A limitation with this approach is that total QA decisions may not directly reflect productivity as the volume of total decisions can reflect unobserved characteristics of the AOI System, limiting the number of images seen by the AI or the human operator.

Panel B of Table 3 reports on the estimation results. The baseline specification shows no statistically significant relationship between changes in AI Automation Rate and total throughput in either the unweighted or weighted specification. This suggests that increasing AI automation neither expands nor contracts the overall volume of QA activity at the week- plant level.

We again test for anticipation of AI deployment by including one- and two-week leads of Δ AI

Automation Rate. The results provide no consistent evidence of anticipation effects. While the two-week lead shows marginal significance in the unweighted specification, this relationship disappears in the weighted specification and the one-week lead is insignificant in both.

Overall, we find no systematic relationship between AI deployment and total QA volume, suggesting that automation is associated with reallocation from operators to AI, rather than leading to increase in the number of QA decisions taken. Thus, shifts in labour demand are moderated by diffusion processes rather than productivity effects.

4.4 Total and Operator OK Rate

Next, we turn to the relationship between AI Automation and the outcomes of the QA process. While we are unable to detect a relationship between AI automation and total QA volume, productivity could be affected by making the QA process more efficient. To test this relationship, we estimate 1 using Δ OK Rate as the outcome variable. Panel C of 3 reports on the estimation results for both the Overall OK Rate and Operator OK Rate. We are able to detect a small but significant increase in the Total OK Rate of 0.08 percentage points in unweighted specification and 0.04 in the weighted specification, whilst we are unable to detect any change in the Operator OK Rate.

Interpreting these results, the effect of the increase in the Total OK Rate is somewhat ambiguous. Our data does not allow us to identify a ground truth for the components, therefore an increasing OK Rate could either reflect an increase in true positives or false positives. However, several pieces of evidence help us infer the most likely interpretation. First, operators very rarely overturn OK decisions made by the AI. Using operators as the ground truth, we detect a false positive rate of under 1%, indicating that the AI consistently and accurately detects OK components. Second, discussions with the company revealed that in their own internal assessments AI has a better false positive rate than operators. Thus, this leads us to conclude that the detected increase in the Total OK Rate does not represent an increase in defective components being flagged OK (i.e. false positives), but rather a decrease in OK components flagged as defective (i.e. false negatives). A decrease in false negatives increases the efficiency of the QA process, as fewer products are scrapped due to incorrect flagging. This implies that the AI resulted in better prediction of errors, and hence improvements in the decision making task [32].

5 Robustness Checks/ Sensitivity Analysis

Although our dataset is rich, allowing us to assess the relationship between AI Automation and our outcomes, it also contains features that motivate a number of robustness checks (as described in Section 3.1). First, certain plants contribute disproportionately to the sample. In the weighted regressions, the effects may be driven by the plants with many decisions, whilst the unweighted regressions may be effected by plants with low numbers of decisions. Compounding this, some plant-week observations record extreme swings in the AI Automation Rate, as can be seen in Figure 7, highlighting both swings of $\pm$ 30 percentage points. Second, there are data missing product and decision information throughout. Third, our main estimation approach is conducted at the plant level. This allows us to capture any potential spillover effects, however, detected relationships may be driven by compositional effects rather than changes in AI Automation. To address these concerns, we conduct a series of robustness checks to assess the validity of the results presented in the previous subsection.

5.1 Leaving-One-Plant-Out

First, to ensure that our findings are not driven by any particular plants, we re-estimate Equation (1) leaving out one plant at a time. This allows us to assess whether the estimated relationship between AI adoption and outcomes is not driven by one particular plant. Figure A.3 displays the results from this leave-one-plant-out exercise. The red marker corresponds to the baseline estimate using the full sample, while the grey markers show the coefficients obtained when excluding each plant in turn. We find limited variation in the estimated coefficients across plants. Importantly, no single plant seem to drive the overall relationship between changes in AI Automation and the growth rates of our outcomes. This reinforces the interpretation that the estimated effect reflects a general pattern across plants rather than being specific to any individual production unit. In the Appendix we provide confidence intervals are constructed by inverting the wild cluster restricted bootstrap test, imposing the null and using Rademacher weights with 9,999 replications.

5.2 Sample Restriction

Second, we turn to the effects of our sample selection and issues relating to missing data. Table 4 reports on the results from various sample selections, using the weighted specification from Equation (1). First, in Column 1, we present the results from our preferred specification as shown in Table 3. In Column 2, we remove the largest positive and negative weekly swings of AI Automation. While the relationship with labour demand remains consistent,the effects on OK rate shift. We observe that while the relationship with the overall pass rate is attenuated, it is still positive. Meanwhile, Operator OK Rate becomes negative, this is consistent with findings at the Component Library level (see Section 5.3). In Column 3, we remove weeks where the number of lines missing decisions is at least 25% as large as the total number of decisions. In Column 4, we remove weeks where the number of lines missing product information is at least 25% as large as the total number of decisions. Column 5 excludes weeks meeting both criteria. The last three columns demonstrate that the identified relationships between our outcomes and AI Automation are robust to weeks with missing information.

5.3 Component Library Analysis

To address the concerns relating to compositional effects driving our detected relationships we re-estimate Equation (1) at the component library level. Just under 80% of component libraries record under 200 decisions per week, yet this only corresponds to 8% of decisions with component library information. To reduce the effect of small component libraries at the week level, we construct a monthly panel at the component library level. To reduce the effect of shifts in our AI Automation Rate being driven by small library sizes, we remove all component library-month observations where we detect less than 200 decisions corresponding to 2.5% of total decisions.

Table 5 reports the results using our final estimation sample, applying the library size restrictions listed above. The results at the component library level are broadly consistent with our plant-level findings. In Panel A, we observe a statistically significant negative relationship between changes in AI Automation Rate and operator decisions, with coefficients of approximately -2.07 in both weighted and unweighted specifications. This coefficient is larger in absolute magnitude than the plant-level estimate, but the comparison should not be read as stronger displacement at the component library level. As the estimation sample here is restricted to libraries with sufficient decision volume, which have a higher mean automation rate than the plant-level panel, the difference reflects a higher level of AI Automation rather than a difference in the intensity of task substitution. When we include anticipation leads, the primary estimation remains negative and significant. However, we detect a relationship with future AI

deployment predicting current labour demand further representing the endogeneity of AI deployment timing at the component library level (see Appendix 2.

In Panel B, examining total QA decisions, we find a small positive relationship at the component library level, contrasting with the null result at the plant level. This suggests that at the component level, AI deployment may be associated with modest throughput increases, though these effects do not aggregate to significant plant-level productivity gains. Again, this may reflect the fact that plants are aware of AI deployments and adjust production of a particular product centered around an AI deployment. Turning to the anticipation effect, we see positive correlations between current throughput and concurrent changes in the AI Automation Rate, alongside the 1 and 2 month leads. Due to endogeneity in AI deployment and an unbalanced panel we are unable to detect a robust relationship between QA throughput and changes in the AI Automation Rate and regard these estimates as unreliable.

Lastly, in Panel C, we observe that the Total OK Rate increases by approximately 0.05 percentage points per 1 percentage point increase in AI Automation Rate, consistent with our plant-level findings. However, we now observe a small negative relationship with the Operator OK Rate, in contrast to our results at the plant level. This is potentially an intuitive result, overall we are unable to detect an effect on the composition at the plant level, however, the AI takes a share of the OK decisions leaving operators with a pool of decisions with a higher defect rate. This indicates that as AI Automation progresses, the operator task may change. However, this association also must be viewed with caution as we are unable to test whether operators change their behaviour following an AI deployment, nor whether this is affected by spillovers across products. Taken together, these component library-level results reinforce our main findings while highlighting that the plant-level estimates capture both direct task substitution and compositional shifts across products and components.

6 Conclusion

This paper documents the intra-firm diffusion of an AI system and its implications for labour demand within a high-tech manufacturing setting. Using decision-level data from a multinational electronics manufacturer, we track the deployment of an AI visual inspection tool across 14 plants over 15 months, and we separate the extensive margin of deployment across components or products and the intensity of use once deployed, as measured by the share of decisions automated. Plants expand AI use almost entirely along the extensive margin. The share of decisions taken on components where the AI has been enabled rises from 30% to 70% over our observation window. On components where the AI is enabled, it automates around one third of decisions throughout, and we detect no systematic growth in that share.

The divergence between these two margins has implications for how intra-firm diffusion is understood. Epidemic accounts attribute slow diffusion to the gradual accumulation of information about a technology, which reduces the perceived risk of adoption and raises use over time [38]. Our extensive margin is partially consistent with such an account, as plants progressively enable the AI on further products and component libraries. The intensive margin is not. If experience with the tool reduced uncertainty about its performance, plants would be expected to lower the confidence thresholds at which the AI decides autonomously, or to reduce the rate at which operators verify its output. Neither adjustment appears systematically in our data, and time since deployment explains roughly a third of the variation in usage intensity across plant-week observations. What plants appear to learn from experience is that the tool can be extended to further components, not that it can be trusted with a larger share of the decisions on components it already covers.

Rank effects offer a closer fit. Differences in capability across units generate different returns to adoption, and [39] extend the argument from the adoption decision to the intensity of use within firms.

Our setting suggests one qualification to where those capabilities reside. Plants in our data could not expand deployment on their own authority, because the automotive customers whose products they assemble had to approve the use of the AI on each product. A plant could therefore be technically ready, having trained a model that met the firm's internal accuracy thresholds, and still be unable to extend or intensify deployment. The binding constraint on intra-firm diffusion here lies partly outside the adopting unit, in the requirements of the buyer. Rank effects framed solely in terms of the resources and managerial capabilities of the unit itself will understate the constraints on diffusion in supply chains where liability for product failure is contractually allocated.

Our labour demand results should be read in light of this pattern of deployment. The AI absorbed inspection decisions without enlarging the inspection task. We find no relationship between changes in the AI Automation Rate and total QA volume, so the countervailing productivity effect emphasised in the automation literature is absent in this setting. Operator decisions fall as automation rises, at a rate statistically indistinguishable from the one implied by the accounting relationship between the two measures, which indicates a transfer of decisions within the task rather than displacement beyond it. . We also detect a small increase in the overall pass rate, consistent with fewer non-defective components being incorrectly flagged. Thus, the pace and extent of labour demand shifts in such settings are driven more by the diffusion process, rather than countervailing effects identified in the automation literature [1].

Two implications follow for how AI's labour market effects are anticipated. First, exposure measures and firm-level adoption events both potentially overstate near-term substitution, because they capture what a technology could do and whether a firm has begun using it, not the share of the work it performs. The gap between the two margins we measure is the size of that overstatement in one setting. Second, where the depth of deployment is governed by liability and by customer requirements rather than by technical capability, the pace of substitution is set by contractual and organisational arrangements that respond slowly to improvements in the technology itself. These rank effects deserve more study, on the one hand firm capabilities determine whether they will adopt the technology, but the environment in which the unit operates can determine the intensity with which the technology is used. Policy calibrated to capability will misjudge timing in such settings, and the direction of the error is towards expecting substitution sooner than it arrives.

There are several limitation that bound the conclusions we can draw. First, we lack plant-level measures of size, technological capability, managerial practice, and information regarding the markets served so we cannot test rank effects directly and rely instead on the firm's own account of the differences between its plants. Both gaps are addressable with matched deployment and personnel data, which would allow the margins of deployment to be linked to the workers exposed to them. Second, we observe decisions rather than workers, so we cannot distinguish a reduction in operator hours from a reduction in the decisions each operator processes within an unchanged shift. Establishing whether the task-level substitution we document translates into changes in employment or hours requires personnel data that our setting does not provide. Third, the evidence here comes from a single firm and a single task, both of which may be considered non-standard. The company we study is a large multinational manufacturer supplying a highly regulated market in which failures carry safety consequences, and customers hold approval rights over the technology used to inspect their products. Under these conditions, constraints on capability and approval are likely to bind more tightly than in settings where the costs of an incorrect decision are lower. The absence of growth at the intensive margin may therefore be specific to this environment, and a firm facing weaker external requirements might raise the intensity of use as experience accumulates.

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Figures:



(a) Printed circuit boards (PCBs)

(b) Surface Mount Technology (SMT) Line

Figure 1: Exemplary representations of the production environment

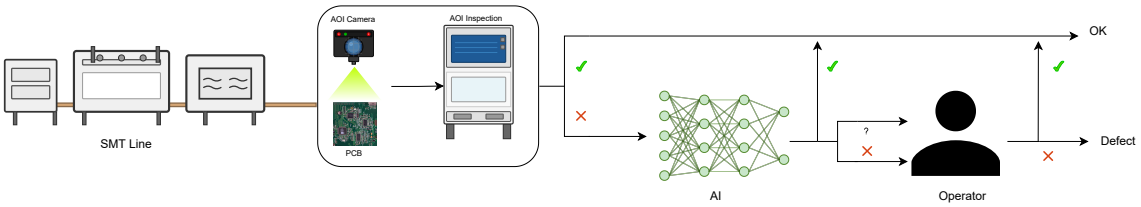


Figure 2: Quality Assurance Process Flow: AOI, AI, and Operator Decision Logic

Notes: This diagram demonstrates the Quality Assurance process flow once the AI has been enabled. The Automated Optical Inspection (AOI) system classifies components as OK or defective, OK decisions are forwarded directly to the back-end. Components classified as defective by AOI are forwarded to the AI system, the AI makes autonomous decisions where it has the required level of confidence (OK components proceed to back-end production). If an image is classified as defective, or the AI is unable to make a decisions, it is forwarded to human operators for final verification. Operator decisions result in either acceptance to back-end production or component rejection.

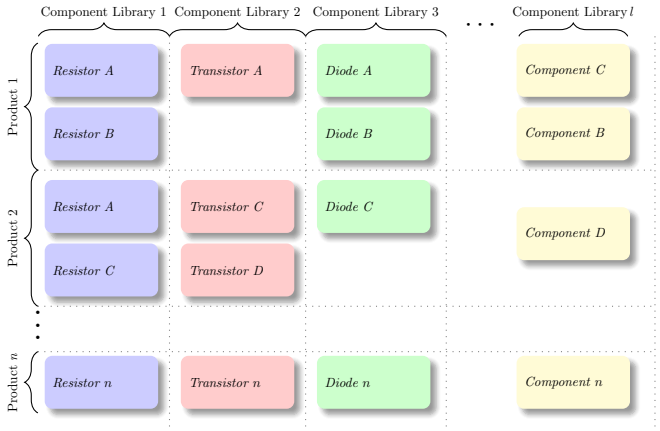
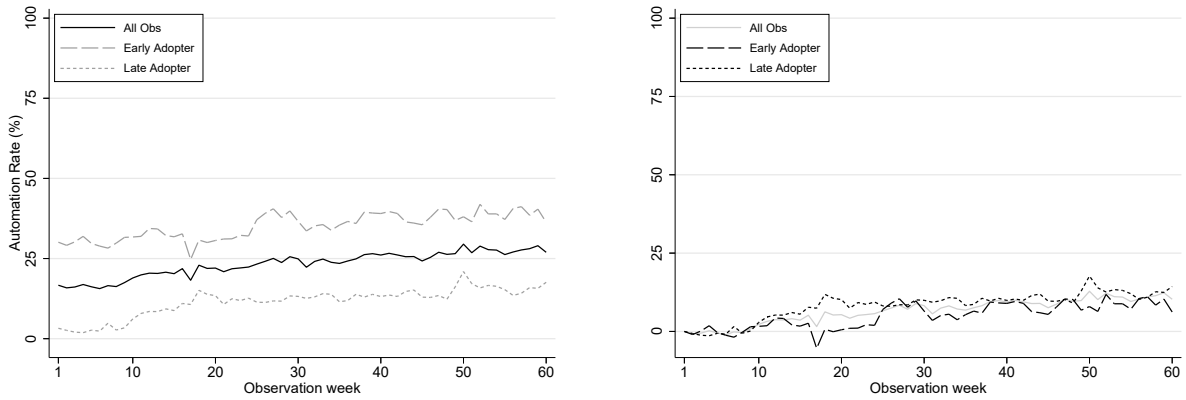


Figure 3: Relationship between Products, Component Libraries, and Components

Notes: This figure displays the representation of our data structure. Components exist at the intersection of Products and Libraries: each product contains multiple components from multiple libraries, and each library contains components appearing across multiple products. This representation shows a fictionalised distribution of components across products and libraries. Component libraries are groupings of similar types of components, here represented as Resistors, Transistors and Diodes. In reality, component libraries will be more granular, representing for example Resistors of a particular size and shape.

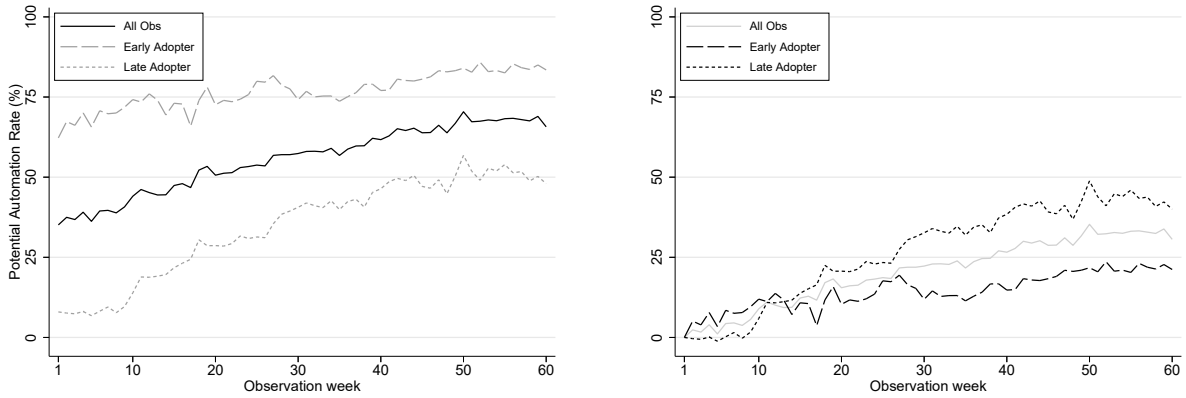


(a) Automation Rate - Levels

(b) Automation Rate -Indexed Growth

Figure 4: AI Automation Rate

Notes: Panel (a) shows the share of QA decisions automated by AI over the 60-week observation period for all plants, early adopters (deployed >8 months before sample start), and late adopters (deployed ≤8 months before sample start). Panel (b) shows the same series indexed to Observation Week 1 to highlight relative growth trajectories.

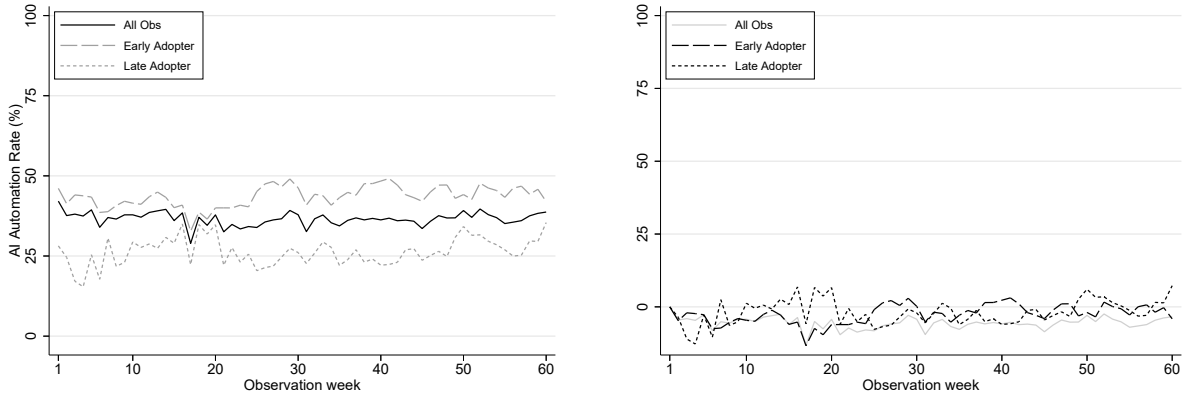


(a) Potential Automation Rate - Levels

(b) Potential Automation Rate Indexed Growth

Figure 5: AI Potential Automation Rate

Notes: Panel (a) shows the share of QA decisions made that could be automated by AI over the 60-week observation period for all plants, early adopters (deployed >8 months before sample start), and late adopters (deployed ≤8 months before sample start). Panel (b) shows the same series indexed to Observation Week 1 to highlight relative growth trajectories.



(a) AI Automation Rate - Levels

(b) AI Automation Rate Indexed Growth

Figure 6: Progression of AI Automation Rate on components where the AI has been deployed

Notes: Panel (a) shows the share of QA decisions automated by AI on components where the AI has been deployed, early adopters (deployed >8 months before sample start), and late adopters (deployed ≤8 months before sample start). Panel (b) shows the same series indexed to Observation Week 1 to highlight relative growth trajectories.

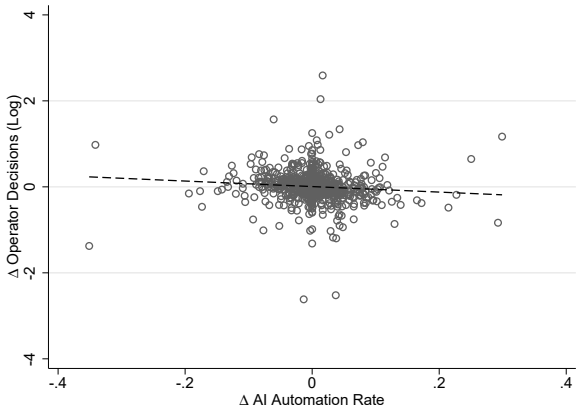
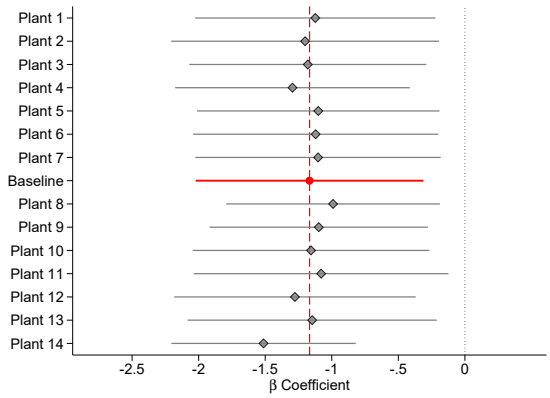
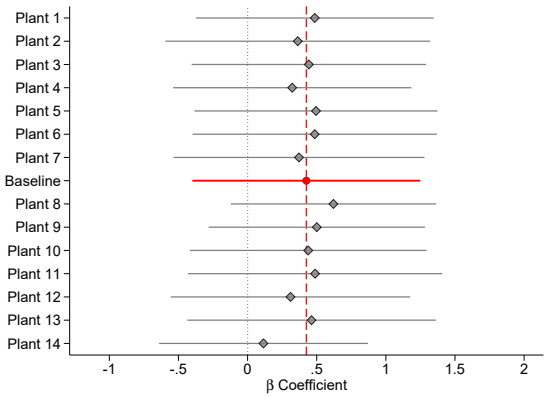


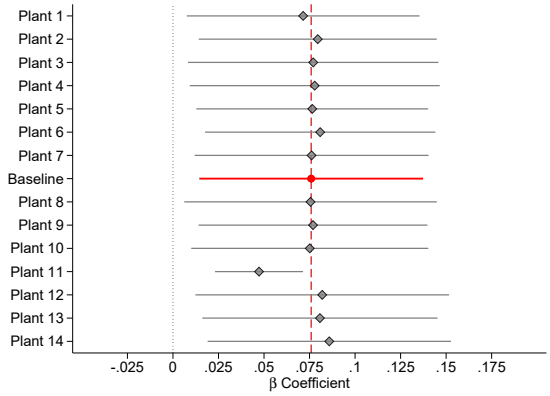
Figure 7: Change in operator decision and AI Automation Rate



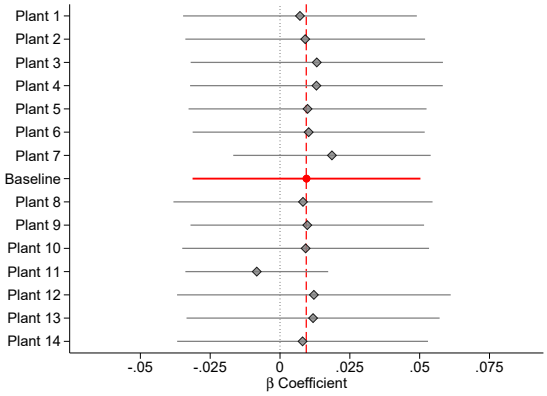
(a) Δ Total Operator Decisions (Log)



(b) Δ Total QA Decisions (Log)



(c) Δ Total OK Rate



(d) Δ Operator OK Rate

Figure 8: Leave-One-Plant-Out Robustness Checks

Notes: Each panel shows coefficient estimates when excluding one plant at a time for each model run including 95% confidence intervals, the Y-axis indicates which plant has been excluded. The red dashed line indicates the baseline estimate using all plants, the black dotted line indicates zero effect. All regressions include plant and week fixed effects with standard errors clustered at the plant level.

Tables:

	Mean	SD	Min	Max
Panel A: Plant Characteristics				
Weeks Observed	59	2	53	60
Unique Products	902	811	130	3303
Unique Component Libraries	1118	671	102	2404
Share of total QA Decisions (%)	7.1	10.6	0.2	42.0
Panel B: AI Automation & QA Outcomes				
AI Automation Rate (%)	23.1	20.6	1.8	67.9
Overall OK Rate (%)	95.3	4.2	86.7	99.8
Operator OK Rate (%)	93.8	5.2	83.4	99.7
Panel C: Week-to-Week Changes				
Δ AI Automation Rate (pp)	0.164	0.109	-0.038	0.374
Δ Total QA Throughput (Log)	0.008	0.018	-0.021	0.046
Δ Total Operator Decisions (Log)	0.006	0.018	-0.021	0.044
Δ Total AI Decisions (Log)	0.030	0.048	-0.024	0.167

Table 1: Summary Statistics

Deployment timing	Number of Plants
Early Adopters (>8 months before)	
>15 months before	4
8-15 months before	3
Late Adopters (<=8 months before/during)	
1-4 months before	2
0-3 months after start	3
8-11 months after start	2
Total number of plants	**14**

Table 2: Plant distribution by AI deployment timing

Table 3: Relationship betwen changes in AI Automation Rate and Main Outcomes

	(1) Unweighted	(2) Weighted
Panel A – Δ Total Operator Decisions (Log)		
Δ AI Automation Rate	-1.201**	-1.216***
	(0.416)	(0.204)
Observations	810	810
Anticipation Effect		
Δ AI Automation Rate (Current)	-1.066**	-1.184***
	(0.414)	(0.208)
Δ AI Automation Rate (lead 1 week)	0.420	0.420*
	(0.264)	(0.186)
Δ AI Automation Rate (lead 2 weeks)	0.442**	0.122
	(0.252)	(0.092)
Observations	800	800
Panel B – Δ Total QA Decisions (Log)		
Δ AI Automation Rate	0.374	0.414
	(0.400)	(0.241)
Observations	810	810
Anticipation Effect		
Δ AI Automation Rate (Current)	0.507	0.442
	(0.401)	(0.254)
Δ AI Automation Rate (lead 1 week)	0.384	0.365
	(0.268)	(0.187)
Δ AI Automation Rate (lead 2 weeks)	0.486**	0.157
	(0.259)	(0.105)
Observations	800	800
Panel C – Δ OK Rate		
Total OK Rate		
Δ AI Automation Rate	0.080***	0.040***
	(0.029)	(0.019)
Observations	810	810
Operator OK Rate		
Δ AI Automation Rate	0.013	-0.015**
	(0.020)	(0.007)
Observations	810	810

Notes: This table presents two-way fixed effects regressions of weekly changes in labor demand (Panel A), total QA throughput (Panel B), and OK Rates (Panel C) on changes in the AI Automation Rate at the week-plant level. The *Anticipation Effect* specification includes one- and two-week leads to test whether future AI deployment correlates with current outcomes. All regressions include plant and week fixed effects. Column (1) reports unweighted estimates; Column (2) weights by total weekly decisions. Standard errors are clustered at the plant level; significance stars are based on wild cluster bootstrap p-values (Rademacher weights, 9999 replications). *** ($p < 0.01$); ** ($p < 0.05$); * ($p < 0.1$).

Table 4: Sensitivity Analysis on Main Outcomes

	(1) Preferred Specification	(2) Outliers	(3) Missing Dec.	(4) Missing Prod.	(5) Missing Prod. & Dec.
Panel A – Δ Total Operator Decisions (Log)					
Δ AI Automation Rate	-1.216***	-1.472***	-1.145***	-1.321***	-1.267***
	(0.204)	(0.231)	(0.224)	(0.189)	(0.179)
Observations	810	728	758	740	702
Panel B – Δ Total QA Decisions (Log)					
Δ AI Automation Rate	0.414	0.208	0.522	0.300	0.390
	(0.241)	(0.279)	(0.250)	(0.223)	(0.209)
Observations	810	728	758	740	702
Panel C – Δ OK Rate					
Total OK Rate					
Δ AI Automation Rate	0.040***	0.022**	0.041**	0.040***	0.041**
	(0.019)	(0.009)	(0.020)	(0.020)	(0.021)
Observations	810	728	758	740	702
Operator OK Rate					
Δ AI Automation Rate	-0.015**	-0.027	-0.016**	-0.015*	-0.016**
	(0.007)	(0.023)	(0.008)	(0.007)	(0.008)
Observations	810	728	758	740	702

Notes: This table presents weighted two-way fixed effects regressions of weekly changes in labor demand (Panel A), total QA throughput (Panel B), and OK Rates (Panel C) on changes in the AI Automation Rate at the week-plant level with various sample restrictions applied. Column (1) reports the weighted estimates reported in Table 3; Column (2) removes the top and bottom 5% of changes in AI Automation Rate; Column (3) removes weeks with a high proportion of lines missing decisions; Column (4) removes weeks with a high proportion of missing products; Column (5) removes both. Standard errors are clustered at the plant level; significance stars are based on wild cluster bootstrap p-values (Rademacher weights, 9999 replications). All regressions include plant and week fixed effects. *** ($p < 0.01$); ** ($p < 0.05$); * ($p < 0.1$).

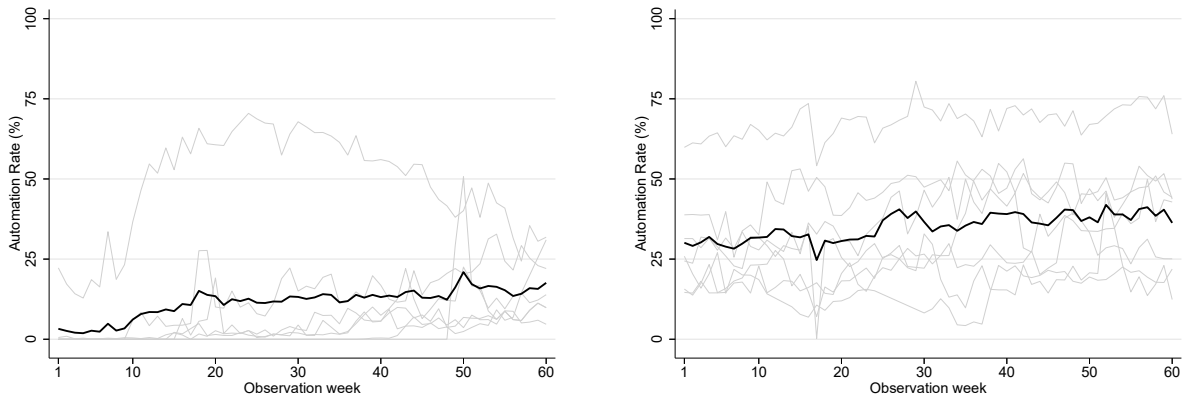
Table 5: Relationship between changes in AI Automation Rate and Main Outcomes using Component Library Panel

	(1) Unweighted	(2) Weighted
Δ AI Automation Rate	-2.066***	-2.071***
	(0.067)	(0.068)
Observations	24361	24361
Anticipation Effect		
Δ AI Automation Rate (Current)	-1.920***	-1.936***
	(0.075)	(0.075)
Δ AI Automation Rate (lead 1 month)	0.125***	0.142***
	(0.046)	(0.046)
Δ AI Automation Rate (lead 2 months)	0.206***	0.218***
	(0.043)	(0.043)
Observations	18750	18750
Panel B – Δ Total QA Decisions (Log)		
Δ AI Automation Rate	0.282***	0.277***
	(0.062)	(0.063)
Observations	24598	24598
Anticipation Effect		
Δ AI Automation Rate (Current)	0.373***	0.360***
	(0.068)	(0.069)
Δ AI Automation Rate (lead 1 month)	0.115***	0.126***
	(0.043)	(0.044)
Δ AI Automation Rate (lead 2 months)	0.190***	0.200***
	(0.042)	(0.042)
Observations	18909	18909
Panel C – Δ OK Rate		
Total OK Rate		
Δ AI Automation Rate	0.049***	0.048***
	(0.006)	(0.006)
Observations	24598	24598
Operator OK Rate		
Δ AI Automation Rate	-0.028***	-0.028***
	(0.007)	(0.006)
Observations	24361	24361

Notes: This table presents two-way fixed effects regressions of monthly changes in labor demand (Panel A), total QA throughput (Panel B), and OK Rates (Panel C) on changes in the AI Automation Rate at the month-component library level. The *Anticipation Effect* specification includes one- and two-month leads to test whether future AI deployment correlates with current outcomes. All regressions include component library and month fixed effects. Column (1) reports unweighted estimates; Column (2) weights by total monthly decisions. Standard errors are clustered at the plant level. *** ($p < 0.01$); ** ($p < 0.05$); * ($p < 0.1$).

Online Appendix

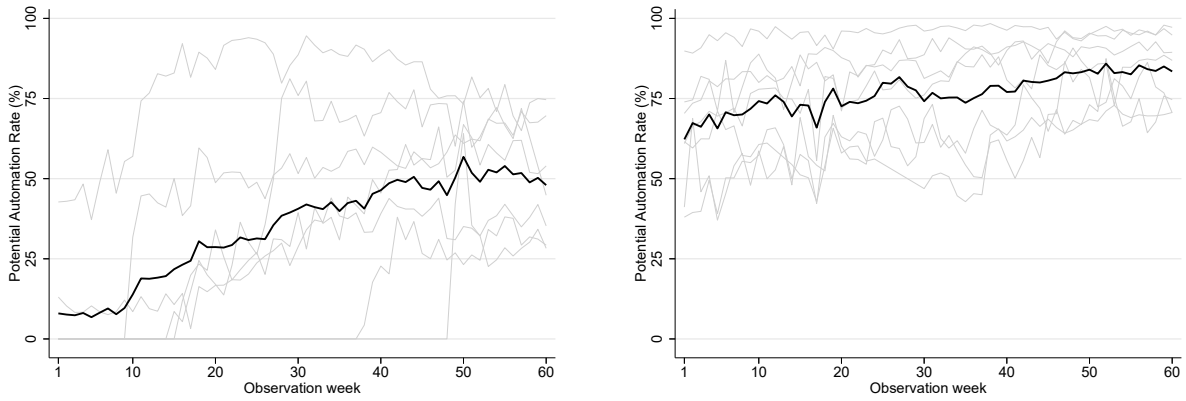
1 Appendix Figures



(a) Late Adopters (b) Early Adopters

Figure A.1: Plant-Level Automation Rate Heterogeneity by Adopter Group

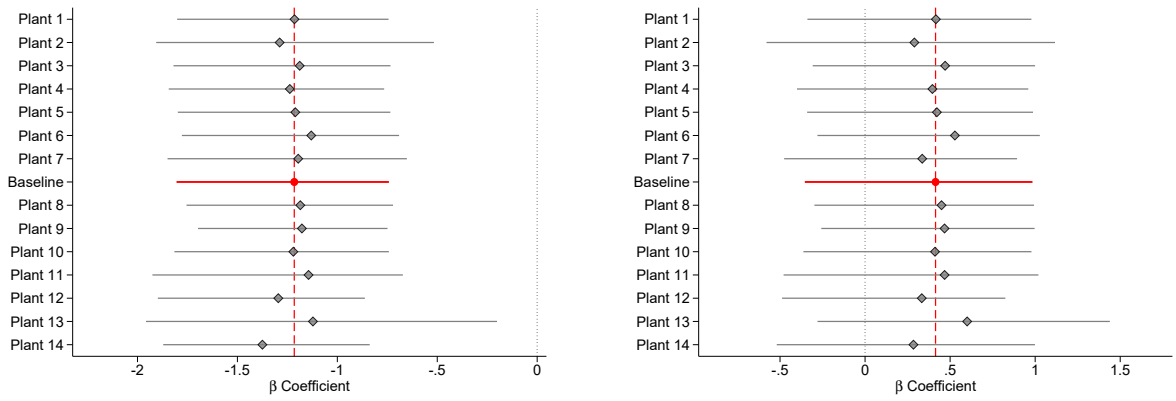
*Note:*Individual plant trajectories (thin gray lines) and group mean (thick black line) for automation rates over the 60-week observation period. Panel (a) shows late adopters (N=7 plants, deployed ≤8 months before sample start). Panel (b) shows early adopters (N=7 plants, deployed >8 months before sample start).



(a) Late Adopters (b) Early Adopters

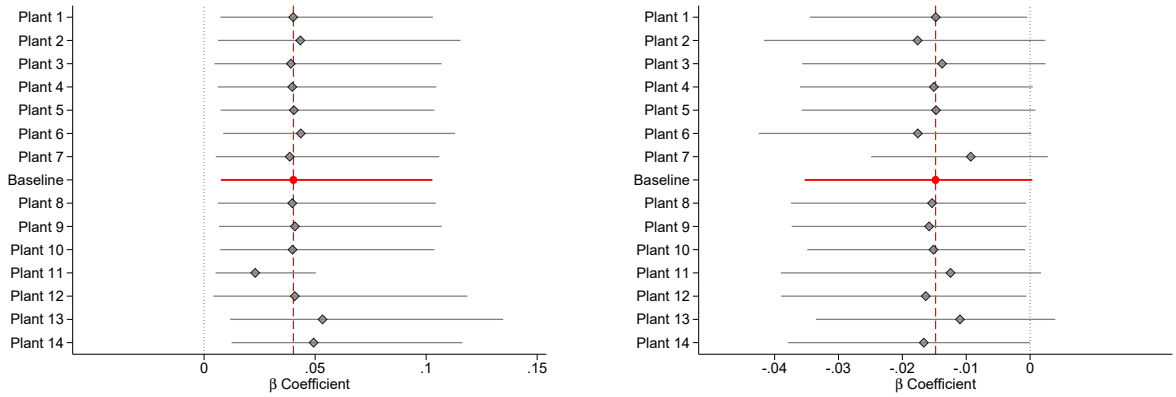
Figure A.2: Plant-Level Potential Automation Rate Heterogeneity by Adopter Group

*Note:*Individual plant trajectories (thin gray lines) and group mean (thick black line) for potential automation rates over the 60-week observation period. Panel (a) shows late adopters (N=7 plants, deployed ≤8 months before sample start). Panel (b) shows early adopters (N=7 plants, deployed >8 months before sample start).



(a) Δ Total Operator Decisions (Log)

(b) Δ Total QA Decisions (Log)



(c) Δ Total OK Rate

(d) Δ Operator OK Rate

Figure A.3: Leave-One-Plant-Out Robustness Checks

Notes: Each panel shows coefficient estimates when excluding one plant at a time, the Y-axis indicates which plant has been excluded. The red dashed line indicates the baseline estimate using all plants, the black dotted line indicates zero effect. All regressions include plant and week fixed effects with standard errors clustered at the plant level. Intervals are 95% confidence intervals obtained by inverting the wild cluster restricted bootstrap test, imposing the null with Rademacher weights and 9,999 replications; they are not symmetric around the point estimate.

2 Identification Challenges

There are a number of factors that limit causal identification in our setup. At more granular levels than the plant month we encounter violations of the Stable Unit Treatment Value Assumption (SUTVA). Consider deployment at the component library level, where treatment is defined as whether AI has been approved for any product containing a component from that library. Because individual products contain components from multiple libraries, spillovers are inevitable. For example, if a PCB contains both a resistor (with AI deployed) and a diode (without AI deployed), AI adoption for the resistor affects outcomes for the diode as they are produced on the same production line. Operator workload may decrease due to reduced labour demands, but this could be offset by faster production times. They may also face increased cognitive demand due to a higher proportion of defective images, impacting their decision making on the diode. If we assigned treatment at the product level, we would be able to rule out this possibility. However, operators work across multiple lines. As we are unable to identify the lines on which operators work, we are unable to rule out the fact that decisions on untreated products may be affected by treated ones. Imagine a factory with two lines, each producing separate products. One product has the AI enabled, the other does not. An operator may be making decisions on products from

both lines, thus the AI automating a proportion of decisions from one line could affect decisions made on products from another.

Aside from the SUTVA violations, we also encounter problems relating to the endogeneity of treatment timing. Plants do not deploy AI randomly across products or component libraries. As described in Section 2.5, deployment requires customer approval, trained models with sufficient labeled data, and plant-level confidence in the technology's performance. Critically, plants may strategically time AI deployment to coincide with increases in production volume associated with new orders or high-volume production periods. Figure A.1 illustrates this problem using the estimator of de [41] to trace the evolution of total QA decisions around AI deployment at the component library level. Total decisions increase sharply in the period leading up to and immediately after deployment, rising by approximately 50% in the month of deployment. This spike demonstrates that deployment timing is correlated with production shocks rather than being exogenous. Because we cannot distinguish whether changes in operator decisions reflect genuine task displacement or compositional shifts in the product mix being inspected, we cannot construct a credible counterfactual for labor demand at the product or component library level, even with staggered difference-in-differences estimators designed to handle dynamic treatment effects.

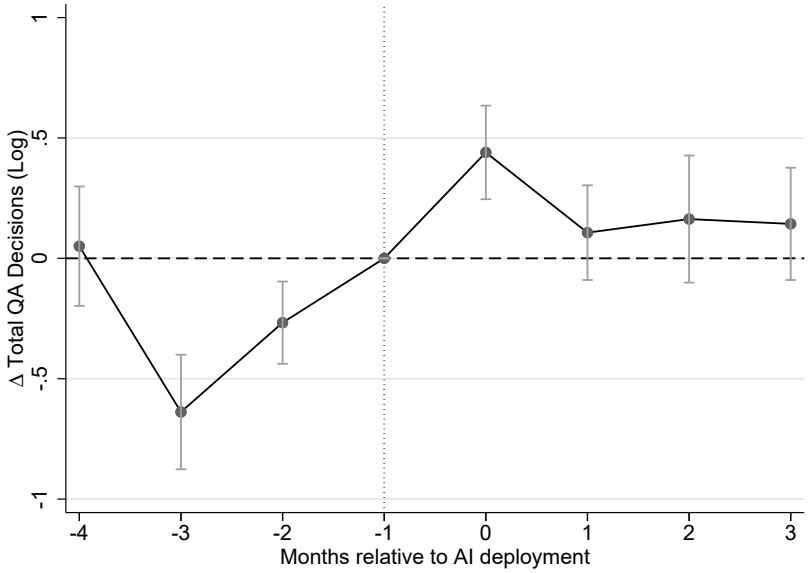


Figure A.1: Endogeneity of AI Deployment Timing

Notes: This figure shows coefficients from the [41] estimator at the component library level. The vertical line indicates AI deployment (t=1). Total QA decisions increase sharply following deployment, rising by approximately 50% at t=1. This spike demonstrates that deployment timing coincides with production increases, preventing causal identification at this level of aggregation.